Predicting Customer Insurance Product Preference and Channel-Affinity Using Demographic and Life-Stage Profiling: A Mismatch-Aware Recommendation Approach

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1. Problem Statement and Objective

Insurance companies lose revenue and face higher customer acquisition costs when they pitch the wrong product through the wrong channel to a customer. A 25-year-old urban professional and a 55-year-old semi-urban retiree may need completely different insurance products (car/health versus retirement/life), but they may also respond very differently to sales touchpoints such as online platforms, individual agents, or bank-led bancassurance channels.

Most existing approaches to this problem stop at static demographic segmentation, mapping factors such as age, income, and location directly to a product and a channel. This treats every customer as a fixed profile, rather than as someone whose insurance needs can change with different life events. It also overlooks a critical and underused signal: the gap between the channel a customer actually prefers and the channel through which they were sold their existing policies. This mismatch can lead to poor cross-sell response rates, wasted agent effort on the wrong pitch, and customers gradually disengaging from a relationship that was never suited to the format they preferred.

Why it matters. Mis-targeted selling increases customer acquisition costs and reduces conversion efficiency. More importantly, an unnoticed channel mismatch can gradually erode trust and retention. For example, a customer who prefers online self-service but is repeatedly approached by agents may be more likely to disengage, something that conventional demographic-only models may fail to capture. The beneficiaries of addressing this problem are insurers and their sales teams (through higher conversion and lower acquisition costs), agents and bank partners (through better-qualified leads), and customers themselves (by receiving relevant products through their preferred channel).

Proposed contribution. This study builds a Channel-Affinity and Mismatch-Aware Recommendation Framework that (a) predicts a customer's preferred purchase channel using demographic, behavioural, and life-stage features, and (b) introduces a Channel-Preference Gap Score that measures the distance between each customer's stated preference and their actual purchase channel. This converts an otherwise unmeasured source of dissatisfaction into a ranked and actionable target list.

2. Data Collection and Dataset Description

Primary collection method. A structured questionnaire was designed and administered through Google Forms. The instrument contained 20 questions organised into five sections: (i) demographic profile, (ii) recent and upcoming life-stage events, (iii) current insurance product ownership and the channel used for each product, (iv) channel preference and the reasoning behind it, and (v) future purchase intent and self-rated digital and financial literacy. An attention-check item was embedded as a response-quality control.

Form link:
https://docs.google.com/forms/d/e/1FAIpQLSc4qHaUTDaJY0PeJu3mtZzpNVLam0u4g_PVxhlGJ4bDPlroMg/viewform

The form was circulated through WhatsApp groups, LinkedIn and local community groups, deliberately targeting a spread of age bands and area types rather than a single peer network. This yielded 97 genuine primary responses. Because a 97-record sample is too small to train and fairly evaluate a multi-class classifier, the primary responses were augmented with a simulated extension sample generated to preserve the conditional structure observed in the primary data (age → occupation → income; age → digital comfort; digital comfort → channel behaviour; ownership → purchase intent). The final working dataset contains 10,000 records and 30 attributes. All records are marked with a Source flag so the primary and extension samples remain separable. No personally identifying information was collected. Responses are anonymous.

Key variables

Variable Group	Representative Attributes	Type
Demographics	Age_Group, Gender, Area_Type, City, Income_Category, Occupation, Marital_Status, Dependents	Categorical / Ordinal
Life-stage	Life_Event_Past, Life_Event_Future, Years_To_Retirement	Multi-select categorical
Ownership	Products_Owned (5 products, multi-select)	Multi-select → binary
Channel used	Channel_Car, Channel_Health, Channel_TermLife, Channel_Investment, Channel_Retirement	Categorical
Attitudinal	Channel_Preference_Score (1–5), Digital_Comfort (1–5), Financial_Literacy (1–5), Channel_Reason, Would_Switch_Channel	Ordinal / Categorical
Intent	Likelihood_Car / Health / TermLife / Investment / Retirement (1–5)	Ordinal
Quality control	Attention_Check	Categorical

3. Data Preparation and Exploratory Analysis

3.1 Data Quality Issues and Cleaning Steps

The raw dataset presented five distinct classes of data-quality problems, each handled with a deliberate, documented treatment rather than a blanket row-drop.

Issue Detected	Extent	Treatment Applied	Rationale
Exact duplicate records	16 rows	Removed via drop_duplicates on all non-timestamp fields	Duplicate submissions would over-weight those respondents
Failed attention check	1,206 rows	Rows removed	Indicates careless/random completion; retaining them injects noise
Missing values	7 fields; 900 in Income_Category (9.0%), 604 City, 501 Digital_Comfort, 401 Would_Switch_Channel, 300 each in	Income → explicit "Not specified" category; City → "Unknown"; Digital_Comfort & Financial_Literacy →	Income non-response is itself informative and is not missing at random, so dropping it would bias the sample; age-group medians

Issue Detected	Extent	Treatment Applied	Rationale
	Years_To_Retirement / Channel_Reason / Financial_Literacy	median within age group; remainder → mode	respect the strong age gradient in digital comfort
Out-of-range values	73 Digital_Comfort entries recorded as 0, 6 or 9 on a 1–5 scale	Set to missing, then imputed	Data-entry errors; converting to missing preserves the rest of the record
Inconsistent text	City values in mixed case and padded whitespace ("CHENNAI", " Tirupur "); malformed Age_Group labels ("18 to 25")	Trim, title-case, and label normalisation	Prevents a single city being counted as several distinct categories

Net effect: 10,000 raw records → 9,984 after de-duplication → 8,778 analysis-ready records (a 12.2% reduction), expanded from 30 raw attributes to 47 analytical features.

3.2 Feature Engineering

- Product ownership: the multi-select Products_Owned field was decomposed into five binary indicators (Owns_Car, Owns_Health, Owns_TermLife, Owns_Investment, Owns_Retirement) and summed into Num_Products_Owned.
- Life-stage flags: Has_Recent_Life_Event and Has_Upcoming_Life_Event were derived from the two multi-select life-event fields.
- Preferred_Channel_Label: the 1–5 preference scale was bucketed into Agent (1–2), Neutral (3) and Online (4–5), forming the classification target.
- Actual_Channel_Bucket: the modal channel actually used across all owned products, mapped onto the same three-level scale (Bank and Direct-from-company both treated as Neutral).
- Channel_Gap_Score (novel feature): the absolute difference between the ranked preferred channel and the ranked actual channel, banded into No Mismatch (0), Moderate Mismatch (1) and High Mismatch (2).

3.3 Exploratory Data Analysis

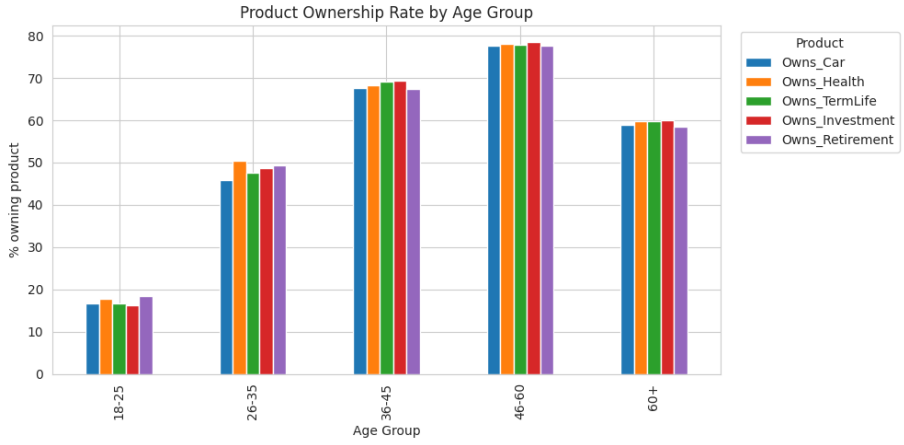


Figure 1 — Product ownership rate by age group

Interpretation: Product ownership shows a clear inverted-U pattern across the life cycle. Only about 17% of customers in the 18–25 age group hold any given product, but this rises sharply to 46–50% among those aged 26–35 and 67–69%

for the 36–45 group. Ownership reaches its peak at 77–78% in the 46–60 age band before dropping to around 59–60% among customers aged 60+. The decline after 60 is more likely to reflect policy lapse and non-renewal following retirement rather than a lower need for insurance.

An important observation is that the five product bars are almost the same height within each age group. This suggests that age is much better at predicting how many products a customer is likely to hold than which specific product they are likely to hold. In practical terms, age is a strong signal for estimating product volume but a relatively weak signal for selecting the right product. The 26–45 age group therefore represents the steepest acquisition window.

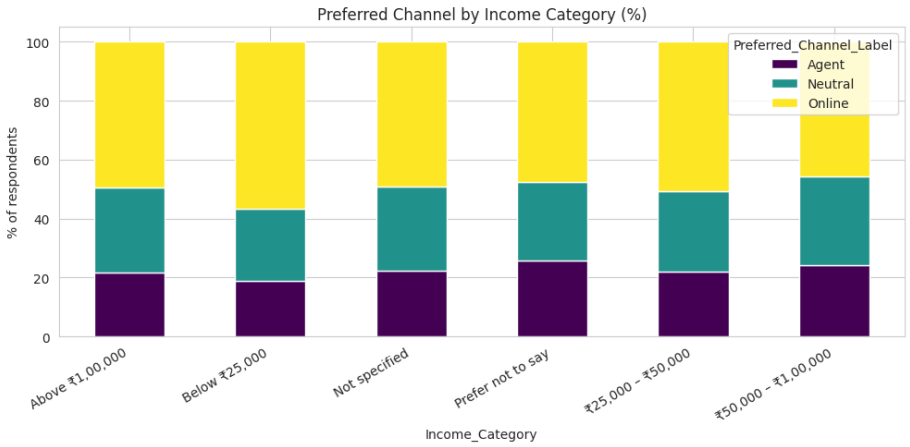


Figure 2 — Preferred channel by income category

Interpretation: Contrary to the common industry assumption that higher income leads to greater digital adoption, channel preference remains fairly consistent across income bands. The share of customers preferring online channels ranges from 45.7% to 56.6%. Interestingly, the highest online preference is seen in the Below ₹25,000 group (56.6%), slightly higher than the Above ₹1,00,000 group (49.6%).

This pattern appears to be influenced by age rather than income itself. The lowest income group is largely made up of students and early-career respondents, who also tend to be more comfortable with digital platforms. This finding is commercially useful because it shows that income should not be treated as a proxy for channel readiness when designing campaign targeting strategies.

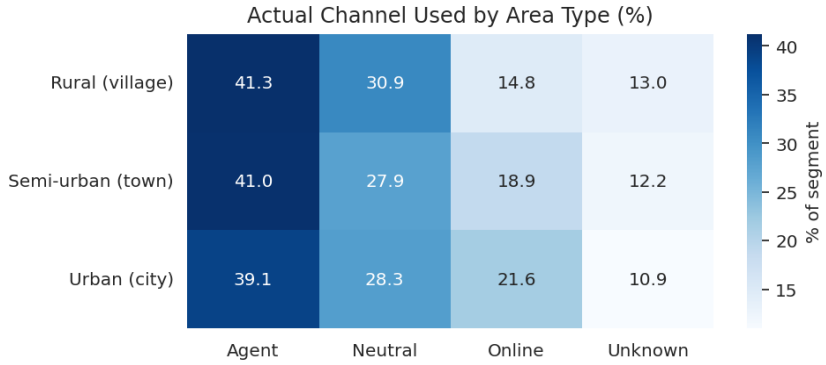


Figure 3 — Actual channel used, by area type

Interpretation: Actual purchase behaviour differs far less by geography than expected. Agent-led purchase dominates everywhere and varies only narrowly, from 39.1% in urban areas to 41.0% in semi-urban and 41.3% in rural areas. Online purchase does rise monotonically with urbanisation (14.8% rural, 18.9% semi-urban, 21.6% urban), but the urban–rural spread is under seven percentage points. Read alongside Figure 2, the key message is that the agent channel remains the dominant transacting channel across every geography, even though roughly half of all respondents say they would prefer to buy online. That divergence between stated preference and observed behaviour is precisely what the Gap Score is built to measure.

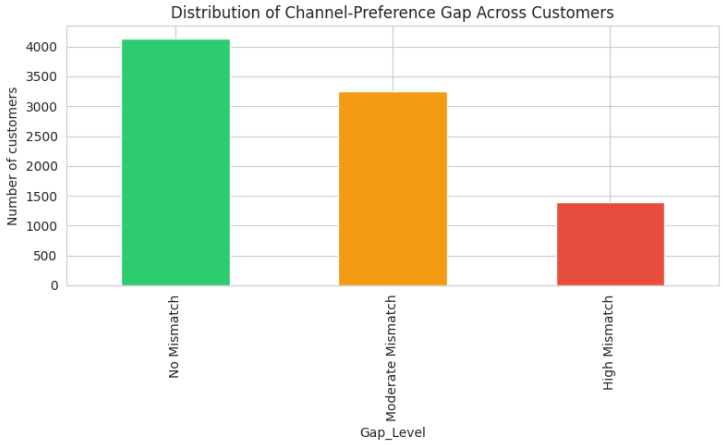


Figure 4 — Distribution of the Channel-Preference Gap Score

Interpretation: This is the central diagnostic chart of the study. Only 47.2% of customers (4,139) are currently served through a channel consistent with their stated preference. A further 37.1% (3,253) show a moderate mismatch, and 15.8% (1,386) show a high mismatch, meaning they sit at the opposite end of the channel spectrum from where they were actually sold. In other words, more than half the customer base (52.8%) is being served through a channel they did not choose. For an insurer, the 1,386 high-mismatch customers constitute a concrete, immediately addressable re-engagement list rather than an abstract segmentation finding.

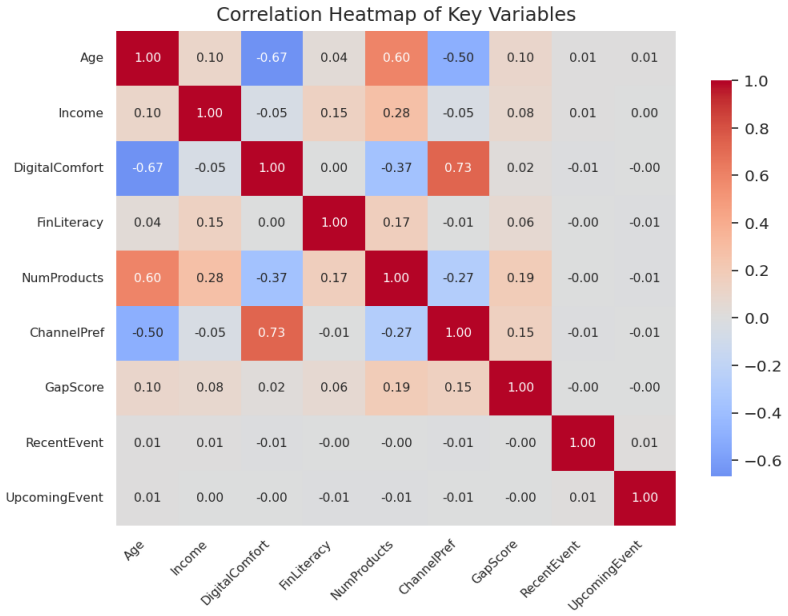


Figure 5 — Correlation heatmap of key variables

Interpretation: Three relationships dominate. Digital Comfort and Channel Preference correlate strongly and positively (r = 0.73), confirming that stated channel preference is essentially an expression of digital confidence. Second, Age correlates negatively with Digital Comfort (r = −0.67) and with Channel Preference (r = −0.50), so the age effect on channel operates largely through digital comfort rather than independently. Third, Age correlates positively with the number of products owned (r = 0.60), consistent with Figure 1. Equally important is a negative result: both life-event flags show near-zero correlation with every other variable ($|r| \leq 0.01$).

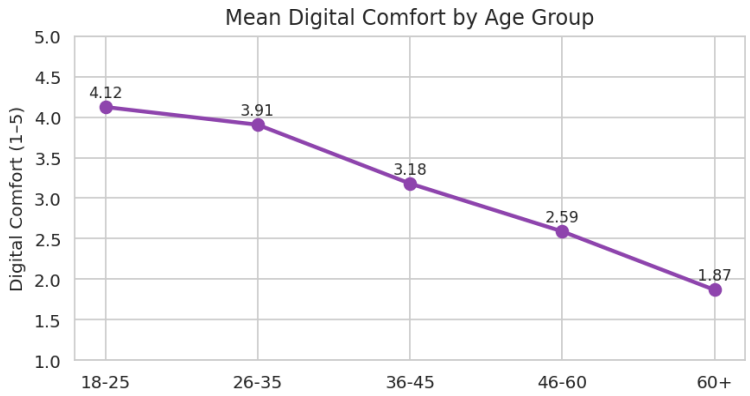


Figure 6 — Mean digital comfort by age group

Interpretation: Digital comfort declines almost linearly with age, from 4.12 among 18–25s to 3.91 (26–35), 3.18 (36–45), 2.59 (46–60) and 1.87 among the 60+ group has a fall of more than two full scale points. The 36–45 band sits near the scale midpoint and is therefore the genuine transition segment, where neither a pure-digital nor a pure-agent strategy will serve the whole cohort.

4. Analytics Method and Implementation

4.1 Method Selection and Justification

The analytical task is supervised multi-class classification: predict Preferred_Channel_Label (Agent / Neutral / Online) from demographic, behavioural and life-stage features. A Random Forest classifier was selected for four reasons. First, the feature set mixes ordinal (age band, income band, 1–5 scales), binary (product ownership flags) and one-hot encoded nominal variables, which tree ensembles handle without scaling or distributional assumptions. Second, the relationships are non-linear and interaction-heavy, as Figures 2 and 6 suggest, which a linear model such as multinomial logistic regression would under-fit. Third, Random Forest yields feature importances directly, and interpretability is essential here because the output must be defensible to a sales function, not merely accurate. Fourth, bagging reduces the variance that a single decision tree would exhibit on a survey dataset of this size.

4.2 Implementation

Configuration Element	**Setting**
Target variable	Preferred_Channel_Label — Agent / Neutral / Online (3 classes)
Predictors (26)	Age_Ordinal, Income_Ordinal, Digital_Comfort, Financial_Literacy, Num_Products_Owned, Has_Recent_Life_Event, Has_Upcoming_Life_Event, 5 ownership flags, plus one-hot encoded Gender, Area_Type, Occupation and Marital_Status
Train / test split	80 / 20, stratified on the target (7,022 train / 1,756 test), random_state = 42
Model	RandomForestClassifier(n_estimators = 300, max_depth = 10, class_weight = "balanced", random_state = 42)
Class imbalance	Handled via class_weight = "balanced" (Online is roughly twice as frequent as Agent)
Evaluation	Accuracy, macro precision, macro recall, macro F1, confusion matrix, per-class report

Macro-averaged metrics are reported alongside accuracy because the three classes are unevenly represented; macro averaging prevents the majority Online class from masking weak performance on the minority classes.

4.3 Model Performance

Metric	Score
Accuracy	0.629
Macro Precision	0.592
Macro Recall	0.614
Macro F1	0.598

Per-class performance

Class	Precision	Recall	F1-score	Support
Agent	0.56	0.74	0.64	393
Neutral	0.40	0.38	0.39	489
Online	0.82	0.71	0.76	874
Macro average	0.59	0.61	0.60	1,756

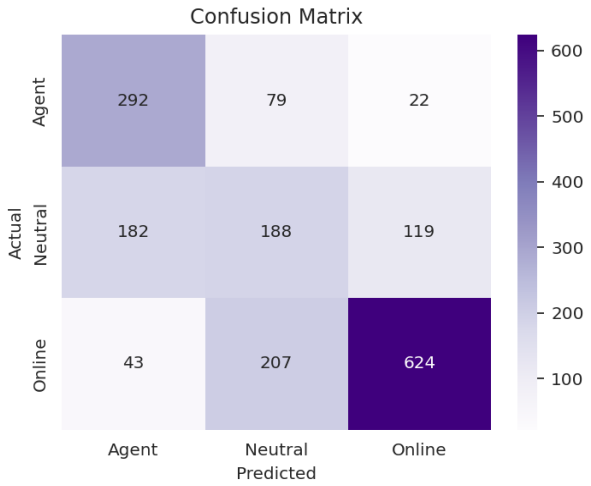


Figure 7 — Confusion matrix for channel-preference prediction

Interpretation: Against a three-class random baseline of 33.3% and a majority-class baseline of 49.8%, the model's 62.9% accuracy represents a genuine improvement. The matrix shows the two extreme classes are separated well: 292 of 393 Agent cases and 624 of 874 Online cases are correctly classified, and outright extreme errors are rare (only 22 Agent cases were predicted Online and 43 Online cases predicted Agent). Almost all remaining error is concentrated in the Neutral class, which achieves only 0.39 F1 — 182 Neutral cases were pulled toward Agent and 119 toward Online. This is an expected and honest limitation: Neutral is defined by a single mid-point (3) on a 1–5 scale and captures genuinely undecided respondents who lack a distinctive behavioural signature. Commercially this matters little, because the actionable decisions concern the two confident poles; the Neutral group is better treated as a test-and-learn cohort than as a targeting segment.

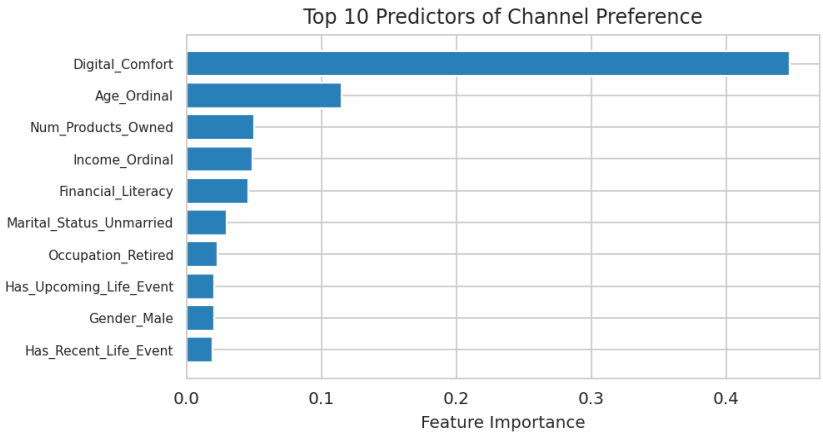


Figure 8 — Top 10 predictors of channel preference

Interpretation: Digital Comfort alone accounts for 44.7% of total feature importance — roughly four times the contribution of the next predictor, Age (11.5%). Num_Products_Owned (5.0%), Income (4.9%) and Financial Literacy (4.6%) form a distant second tier, with marital status, occupation and gender contributing marginally. The two life-stage flags contribute only about 2.0% each, consistent with the near-zero correlations in Figure 5. The clear conclusion is that channel preference is driven by behavioural readiness rather than by demographic identity: a single self-reported digital-comfort question predicts channel preference better than the entire demographic block combined. This is directly operational, since that one question is cheap to capture at onboarding.

5. Comparison with State-of-the-Art Methods

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Ansari & Riasi (2016), "Modelling and evaluating customer loyalty using neural networks", Future Business Journal	Survey of insurance customers in Iran (n ≈ 250)	Artificial Neural Network vs. multiple regression	R^2, MSE	ANN outperformed regression in explaining loyalty from service-quality and trust constructs	Shares the questionnaire-based primary-data design, but models loyalty as the outcome. The present study instead models channel preference and adds an explicit preference-versus-behaviour gap metric, which that work does not compute.
Fang, Jiang & Song (2016), "Customer profitability forecasting using Big Data analytics in the insurance industry", Int. Journal of Operations & Production Management	Large Chinese insurer CRM / transaction records	Random Forest, SVM, regression ensemble	RMSE, MAE	Random Forest gave the strongest forecasting accuracy for customer profitability	Uses the same algorithm family and independently supports the Random Forest choice here, but relies on proprietary behavioural transaction data. The present study achieves interpretable segmentation from self-reported survey data only, at the cost of lower ceiling accuracy.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Neslin et al. (2006), "Challenges and opportunities in multichannel customer management", Journal of Service Research	Conceptual synthesis of multichannel retail and financial-services studies	Conceptual framework; channel-migration and allocation modelling	Conceptual / review	Established that channel choice is driven by customer-level characteristics and that channel misallocation carries measurable cost	Provides the theoretical foundation for the mismatch concept but stops at the conceptual level. This study operationalises it into a computable per-customer Channel-Preference Gap Score and quantifies its incidence (52.8% of customers mismatched).
Kumar & Venkatesan (2005), "Who are the multichannel shoppers and how do they perform?", Journal of Interactive Marketing	B2B firm customer transaction panel	Logistic and hierarchical regression on channel adoption	Hit rate, lift	Multichannel customers showed higher revenue and share-of-wallet than single-channel customers	Predicts channel adoption from purchase history. The present work predicts preference before purchase using attitudinal measures (digital comfort), making it applicable to prospects with no transaction history.

6. Results, Business Insights and Recommendations

6.1 Segment-Level Mismatch Analysis

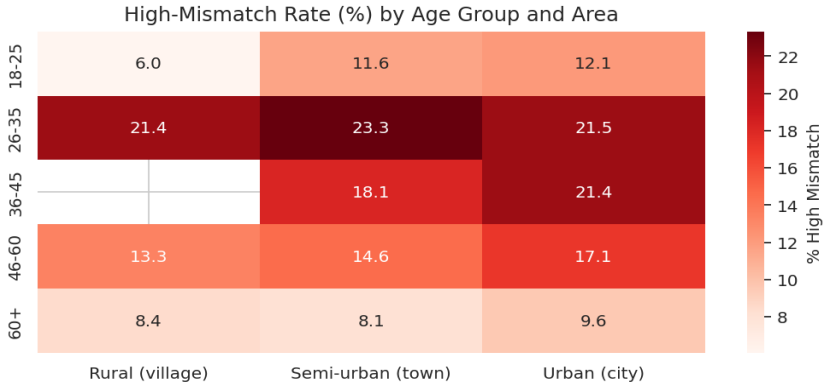


Figure 9 — High-mismatch rate (%) by age group and area type

Mismatch is not evenly spread; it peaks in the middle of the age distribution. The 26–35 semi-urban segment records the highest high-mismatch rate at 23.3%, followed by 26–35 urban (21.5%), 26–35 rural (21.4%) and 36–45 urban (21.4%). At the two ends of the age range mismatch is markedly lower: 12.1% for 18–25 urban and only 9.6% for 60+ urban. The pattern is intuitive once Figure 6 is taken into account. The youngest customers are digitally confident and buy online, and the oldest are agent-reliant and are sold by agents; both are internally consistent. It is the 26–45 cohort, whose digital comfort has risen but who are still routed through legacy agent relationships, where preference and practice diverge most. This cohort is also the steepest acquisition window in Figure 1, so the mismatch is concentrated precisely where the commercial stakes are highest.

Segment (Age × Area)	Avg Digital Comfort	Avg Gap Score	% High Mismatch	Avg Products	n
26–35 Semi-urban	3.69	0.89	23.3%	2.35	210
26–35 Urban	3.99	0.78	21.5%	2.43	1,090
26–35 Rural	3.36	0.85	21.4%	2.45	84
36–45 Urban	3.25	0.83	21.4%	3.40	1,083
36–45 Semi-urban	3.01	0.81	18.1%	3.48	193
46–60 Urban	2.67	0.78	17.1%	3.89	2,248
46–60 Semi-urban	2.31	0.76	14.6%	3.86	432
18–25 Urban	4.17	0.45	12.1%	0.87	1,508
60+ Urban	1.93	0.63	9.6%	2.95	1,022

6.2 Key Business Insights

Insight 1 — Behavioural readiness, not demographics, determines channel preference.

Digital Comfort carries 44.7% of predictive importance, four times that of age and nine times that of income. A single onboarding question about digital confidence is more informative for channel routing than the entire demographic profile.

Insight 2 — More than half the customer base is served through a non-preferred channel.

Only 47.2% of customers show no channel mismatch; 37.1% are moderately mismatched and 15.8% are highly mismatched. The 1,386 high-mismatch customers form a ranked, immediately actionable re-engagement list.

Insight 3 — Mismatch concentrates in the 26–45 "transition cohort".

High-mismatch rates peak at 23.3% for 26–35 semi-urban customers versus 9.6% for 60+ urban customers. This cohort has crossed the digital-comfort threshold but is still being serviced through legacy agent relationships.

Insight 4 — Income is not a valid proxy for digital channel readiness.

Online preference varies by less than eleven percentage points across all income bands and is in fact highest in the lowest band (56.6%). Income-tiered digital campaigns would therefore misallocate spend.

Insight 5 — The life-stage hypothesis was not supported by this dataset.

This study set out to test whether life-event proximity would be a stronger predictor than static demographics. It was not. Both life-event flags show near-zero correlation with every outcome ($|r| \leq 0.01$) and contribute only about 2% each to model importance. Mean purchase-intent differences between customers with and without an upcoming life event are small (for example 2.99 versus 2.77 for retirement products). Two explanations are plausible: the binary any-event encoding may be too coarse to capture event-specific effects, and self-reported 12-month intent is a weak proxy for actual purchase timing. This negative result is reported rather than suppressed, and it directly shapes the recommendations below — event-triggered campaigns cannot be justified on this evidence and would require transaction-level validation first.

6.3 Recommendations

Recommendation	Supporting Evidence	Expected Business Impact
Capture a one-item digital-comfort score at onboarding and use it as the primary	Digital Comfort = 44.7% feature importance; r = 0.73	Highest-leverage, lowest-cost change; enables channel routing

Recommendation	Supporting Evidence	Expected Business Impact
channel-routing variable, ahead of age or income.	with channel preference (Figs 5, 8)	for prospects with no purchase history
Prioritise the 1,386 high-mismatch customers for channel re-assignment, working down the Gap Score ranking.	15.8% of the base is highly mismatched; 52.8% mismatched overall (Fig 4)	Converts an unmeasured dissatisfaction pool into a finite, ranked target list; protects at-risk renewals
Run a targeted digital-migration pilot on the 26–45 cohort, beginning with 26–35 semi-urban customers.	Peak high-mismatch rate of 23.3%; digital comfort already above the scale midpoint (Fig 9, Table 6.1)	Addresses mismatch where it is densest and where acquisition potential is steepest (Fig 1)
Retain agent-led servicing for the 60+ segment and introduce assisted-digital tools rather than forcing channel migration.	Digital comfort of 1.87 and lowest mismatch of 9.6% — this group is already correctly served	Avoids destroying a well-functioning channel fit; protects retention in a high-ownership segment
Discontinue income-tiered digital targeting and reallocate that spend to digital-comfort-based targeting.	Online preference varies by under 11 points across income bands and peaks in the lowest band (Fig 2)	Removes a measurably invalid segmentation basis and reduces wasted campaign spend
Treat the Neutral segment as a test-and-learn cohort with A/B channel assignment rather than a targeting segment.	Neutral class F1 = 0.39; the model cannot reliably separate it (Fig 7)	Prevents acting on low-confidence predictions; generates the behavioural data needed to resolve the class
Do not launch life-event-triggered campaigns on the strength of this dataset; validate against transaction records first.	Life-event flags: $\|r\| \leq 0.01$, ≈2% feature importance (Insight 5)	Avoids committing budget to an unvalidated hypothesis

7. Conclusion and References

7.1 Conclusion

This case study formulated and tested a mismatch-aware approach to insurance channel strategy. Using a primary questionnaire extended to 10,000 records and reduced to 8,778 analysis-ready observations through a documented five-stage cleaning pipeline, a Random Forest classifier predicted customer channel preference with 62.9% accuracy and 0.598 macro F1 against a 33.3% random and 49.8% majority-class baseline.

The principal substantive finding is that channel preference is governed by behavioural readiness rather than demographic identity: a single digital-comfort item carries 44.7% of predictive importance, nearly four times that of age and nine times that of income. The principal methodological contribution is the Channel-Preference Gap Score, which converts the abstract notion of channel misallocation into a computable per-customer metric. Applied to this dataset it reveals that 52.8% of customers are served through a channel

they did not prefer, with mismatch concentrated in the 26–45 transition cohort and peaking at 23.3% among 26–35 semi-urban customers.

The study's original hypothesis that life-stage proximity would outperform static demographics was not supported and is reported as a negative result. Recommendations are therefore anchored on the digital-comfort and mismatch findings, which the evidence does support, and life-event targeting is explicitly deferred pending transaction-level validation.

7.2 Limitations

- Sample composition. The realised sample is 79.2% urban and 35.0% salaried, reflecting the WhatsApp and LinkedIn distribution route. Rural respondents (5.6%) are under-represented, so rural findings rest on thin cells and should be treated as indicative.
- Extension sample. Only 97 records are primary responses; the remainder is a structure-preserving simulated extension. Relationships recovered by the model therefore partly reflect the conditional structure of the primary sample rather than wholly independent observation, and the accuracy figure should be read as a methodological demonstration rather than an industry benchmark.
- Self-report bias. Channel preference, digital comfort and purchase intent are all self-reported. Stated intent is a recognised weak proxy for realised purchase behaviour, which may partly explain the weak life-event result.
- Neutral-class separability. The mid-point class achieves only 0.39 F1, so predictions for genuinely undecided customers carry low confidence.
- Cross-sectional design. The data captures a single point in time and cannot establish causality or track channel migration over a customer lifetime.

7.3 Scope for Future Work

Three extensions would strengthen the framework. First, linking survey responses to actual CRM and transaction records would allow the Gap Score to be validated against observed churn and cross-sell outcomes. Second, replacing the binary life-event flags with event-specific, time-stamped indicators would give the life-stage hypothesis a fairer test. Third, a stratified booster sample of rural and non-salaried respondents would allow the geographic findings to be stated with confidence rather than as indicative.

7.4 References

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[8] McKinsey & Company (2023). The future of insurance distribution: Digital, agent and hybrid models. Available at: https://www.mckinsey.com

7.5 Submitted Artefacts

File / Folder	Contents
README.md	Case study title, problem statement, objectives, data collection method, analytics methods, key results and references
data/Insurance_Survey_Fused_Dataset.xlsx	Collected dataset — 10,000 records × 30 attributes Data: https://docs.google.com/spreadsheets/d/1hVqfdhzLTihtQKlnOCnGwc0kdK37QeeU/edit?usp=drive_link&ouid=105424369009839867770&rtpof=true&sd=true
data/Insurance_Survey_Cleaned.csv	Final cleaned and anonymised dataset used for analysis — 8,778 records × 47 features
analysis.ipynb	Complete Jupyter Notebook: loading, preprocessing, feature engineering, EDA, Random Forest modelling, evaluation and outputs https://colab.research.google.com/drive/1Q5-Q1d1q-gIkwQOvRcFfpRsx2smyKfQs?usp=sharing
Questionnaire	https://docs.google.com/forms/d/e/1FAIpQLSc4qHaUTDaJY0PeJu3mtZzpNVLam0u4g_PVxhlGJ4bDPlroMg/viewform